

Conceptual Design of Emotional and Pain Expressions of a Virtual Patient in a Virtual Reality Training for Paramedics

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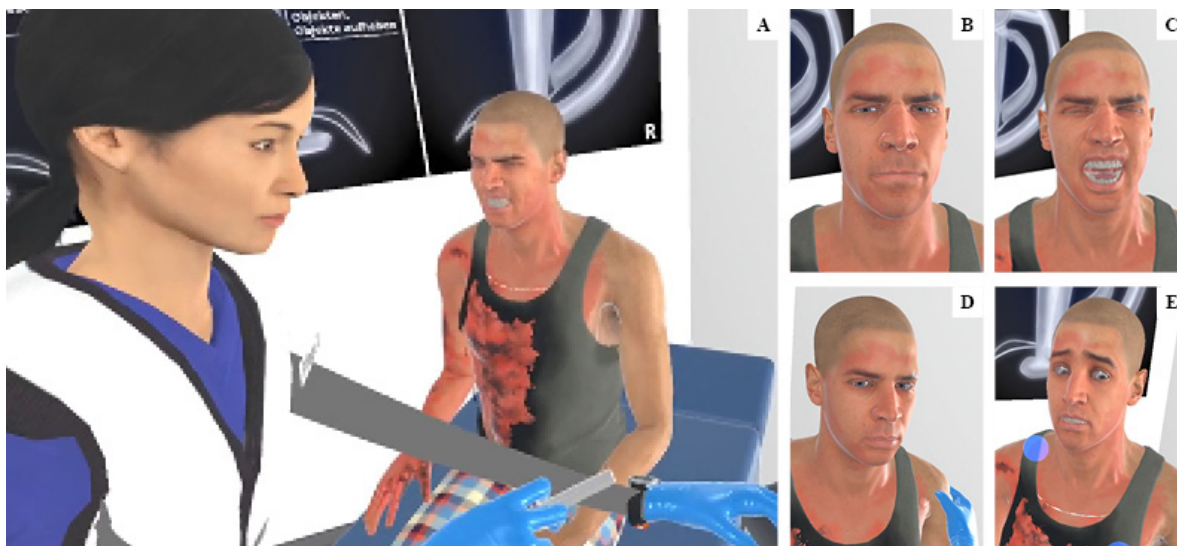


Figure 1: A: VR Setting, B: Pain level 6, C: Pain level 10, D: Relief, E: Fear

ABSTRACT

Incorporating interactive elements into medical virtual reality (VR) training can increase presence and engagement of trainees. Based on this, we designed emotional and pain facial expressions for a virtual patient that interacts with trainees in a VR training scenario for paramedics. We follow a user-centered approach by considering the requirements of trainers. Furthermore, we validate our concept with an online study and we plan a user study with paramedic students as the next step.

Index Terms: Human-centered computing—Interaction design—Contextual design—; Applied computing—Education—Interactive learning environments—

1 INTRODUCTION

Virtual Reality (VR) is being increasingly used in medical training, since it offers the possibility of practicing complex procedures in a safe virtual environment without risky consequences [2, 3, 5, 10]. Although with these modern systems trainees often achieve similar or even better learning effects when compared to more conventional methods, there is still room for improvement [4, 8]. In this paper we analyse the medical and technical requirements to integrate emotional and pain reactions of a virtual patient into a VR training for paramedics. In order to do this, we use mixed methods to get insights from trainers and to validate our interaction concept. With

two studies we develop a concept that will guide the development and design process of the emotional and pain reactions in the VR training.

We start by conducting a focus group with trainers who are experts on prehospital and emergency care in order to explore the added value of including emotions in VR training. Furthermore, we present the results of an online study which guided the design process of pain and emotional expressions of the virtual patient. The developed VR training will be then evaluated in a user study, which goal will be to address the topic of interaction and emotional response of the virtual patient. This paper shows the first user-centered phase that aims at developing an interaction concept for a mixed reality (MR) prototype [12], which focuses on case of a burnt patient, covering prehospital and clinical settings. In the current paper, we focus on the interaction concept for the emotional reaction of the burnt patient. With this paper we present the following contributions:

- Present a mixed methods approach to analyse user requirements (trainers) and validate a development concept for emotional and pain reactions in virtual patients
- An interaction concept that will guide the development of a VR prototype for the training of paramedics
- Set the base for a user study with trainees to assess the success of the developed features

2 RELATED WORK

2.1 Emotional expressions in VR settings

There are many studies and prototypes that integrated complex emotional reactions in training in order to assess the role of these in the

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social aspects of interaction in VR medical training [9]. Having a realistic and adequate emotional response in a medical simulation is important since physicians, paramedics, nurses and other medical staff observe non-verbal cues that alert them about the state of the patient. Precisely, pain expressions and emotional reactions are quite important when interacting with the patient [9]. For instance, in a study of 2008, Kotranza and Lok [7] developed an MR-scenario, where participants (2nd year physician assistants) touched their virtual patients as frequently and as many times as with a standardized patient (trained patient-actor). The virtual patients in this MR scenario were programmed to show pain reactions verbally (e.g., “ouch”, “it hurts”) and visually, by showing pain in her facial expression. The participants reacted accordingly by reporting a higher empathy with patients who show an appropriate emotional response. Precisely, the emotional response of the patient is a relevant feature in any medical training simulation, since medical staff need to interpret signs about the patients’ state by understanding their emotional reactions and pain level in order to act accordingly [9]. Bickmore et al. [1] investigated multimodal interactions (e.g., touch, speech, facial expressions) between humans and a mannequin and found out that facial expressions dominate user perception of affect valence compared to other modalities. They also found out that touch can lead to a better relationship between user and virtual agent. In a study that integrated different emotions of virtual agents in a non-medical VR scenario, Volonte et al. [14] observed the galvanic reaction and questionnaire-responses of their participants when interacting with multiple virtual agents that presented different emotional states. They found that a positive emotional state in virtual agents increased the participants’ number and duration of interactions. This evidence suggests that it is important to design virtual agents with realistic emotional reactions in a medical emergency setting, in order to observe an empathic reaction and increased engagement in training.

3 METHODS

We conducted two studies with different methods and different target groups (trainers and a convenience sample). The goal of this first two studies is to understand the requirements of trainers and the relevance of integrating emotional responses in virtual patients. Furthermore, these studies will be the base for the technical concept that guides the development of the VR training.

3.1 Pre-study: Focus group with trainers

One human-computer interaction researcher and one interaction designer conducted a focus group with trainers of the vocational study for paramedics (2 men), advanced training for emergency nursing (1 woman) and one medical education expert (1 man). The goal of this study is to have an interdisciplinary discussion about the implementation of emotional expressions into the VR training [12]. With this information we aim at understanding educational goals and medical aspects that need to be addressed when developing the VR prototype. Due to the coronavirus pandemic, the focus group was conducted online, using the computer cameras and microphones. No internet problems were identified during the 1 hour discussion. The following research questions served as a guide for the focus group:

- How important are the emotional and pain expressions of patients during a real emergency?
- How important is it to integrate accurate emotional and pain expressions into a VR training?
- In which cases is it important to touch the patient to intensify the social interaction (e.g. to comfort or to reassure)?

3.1.1 Results: Focus group with trainers

All participants consider the emotional and pain expressions as extremely relevant, since these serve as an information source that guide health professionals to focus on relevant aspects during the treatment. The most common emotions in complicated emergencies, especially in burn injuries, are fear, pain, and shame to a minor degree. Pain is usually assessed by asking patients their pain level in a scale from 1 to 10. According to one paramedic, “*this answer is very subjective but it needs to be considered as it comes in order to treat the patient accordingly*”. Including emotional and pain reactions in a VR training is relevant “*as long as the interaction fits educational purposes*”, meaning that a medical procedure should trigger the appropriate pain or emotional response in the patient without delays or extra steps for trainers and trainees.

Physical contact with patients is used in many cases to comfort them and it is usually well received as long as the contact is applied to body parts like shoulders, forearms, and depending on the medical case, even feet or cheeks. With burn patients, the physical contact is particularly challenging, since they might need the comfort provided by touch, but “*health professionals must be careful not to inflict more pain since patients present hypersensitivity and high pain levels*”. There is no professional guide to how and when touch patients. This is a competence mainly learned with experience, which is why it is beneficial to integrate the possibility of touching the patient in the virtual environment, since it might help to get the social competence through this interaction.

3.2 Study: Design of emotions in virtual characters

We performed an online study with 107 participants, (39 women, 63 men, 1 non-binary, 4 with no response) aged between 18 and 63 years old ($M = 29.01$, $SD = 10.42$), that aimed at validating our design of different emotional and pain expressions in the virtual patient. Based on the results of the previous study, we designed 2 emotional expressions, namely fear and shame; 10 different pain expressions ranging from 1 (no pain) to 10 (very high pain), and relief (figure 3). Our design was guided by the Facial Action Coding System (FACS) of Eckmann [11]. The FACS has been used extensively as a facial expression model to simulate emotions in patients, since it allows researchers and designers to change small facial movements and quantify the expressions, which results in the possibility of recreating an emotion in a very precise manner [9]. The online study started with an informed consent, followed by questions about participants’ experience with video games. After this, participants were presented with three tasks:

1. Participants observed the ten facial pain expressions in a pseudo-random order, meaning that the sequence was randomized one time and presented equally to all participants. Their task was to respond in a 4-points Likert scale, ranging from *strongly disagree* to *strongly agree*, whether the facial pain expression they were currently observing was realistic and whether it was credible. The aim of this task was to observe how credible and realistic the participants perceived the facial expressions.
2. Participants observed the 10 facial pain expressions together. Their task was to arrange the pictures from the lowest to the highest pain. Designing realistic facial expressions is a great challenge, even more if the task is to recognize subtle differences and arrange faces in a specific order. Therefore, the goal of this task was to validate our own visual design by contrasting it with the perception of the facial pain expression of participants.
3. Finally, participants saw a video-animation of 22 seconds, in which the virtual character transitioned between the following emotions: neutral, strong pain, relief, fear, mild pain, shame.

Table 1: Arrangement of facial pain expressions.

Facial pain expressions according to the FACS												
	Pain level	1	2	3	4	5	6	7	8	9	10	Total
Facial pain expressions arranged by participants	1	95.3	0.9	2.9	1							100
	2	4.7	81.3	14								100
	3		17.8	78.5	0.9	2.8						100
	4			4.7	60.7	29	0.9	4.7				100
	5				29	42.1	2.8	23.4	0.9	1.9		100
	6				6.5	18.7	37.4	25.2	10.3	0.9	0.9	100
	7				1.9	7.5	28	29.9	26.2	6.5		100
	8						21.5	9.3	43	25.2	0.9	100
	9						6.5	5.6	16.8	64.5	6.5	100
	10						2.8	1.9	2.8	0.9	91.6	100
Total		100	100	100	100	100	100	100	100	100	100	

Their task was to arrange these emotions according to the order they observed on the video. The goal of this task was to test the emotional expressions, while observing if participants can understand and follow the interactive change in emotions on the virtual character, which is a critical feature for the VR prototype.

3.2.1 Results: online study - emotions in virtual characters

In order to test whether the emotions were perceived as believable and realistic, we added the lowest and highest values (*Disagree*= 1 and 2 vs. *Agree*= 3 and 4) to observe how many participants perceived the facial pain expressions as more or less credible and realistic. Figure 2 shows a learning curve. Participants were probably undecided when rating the first pain expressions and scored both aspects on an average level. After the next pictures were shown, participants found the expressions more realistic and believable. This result can be interpreted twofold: the pseudo-random order is accountable for the learning curve that was observed, which obliges us to carefully interpret the results since we might not be testing a clean observation of the realism and credibility of the visual designs. On the other hand, the fact that the continuous presentation of these images has an impact on the participants' perception is a sign that the familiarization with these stimuli might increase the realistic perception in users, which could also increase immersion and involvement in VR. Both interpretations do not exclude one another. However, it would have been beneficial to completely randomize the order in which the pictures were presented.

Regarding the second task, we observe in table 1 that participants arranged the pictures according to our FACS-based design, meaning that the facial pain expressions with their different levels are adequate. Finally, the results of the animation show that participants could not accurately arrange the emotions in the order presented in the animation. The first two emotions, neutral and strong pain, were accurately recognized by 66% and 54% of participants. Similarly, the last emotion, shame, was recognized correctly by 89.7%. The other emotions (relief, fear and mild pain) were confused with each other to different extents. We strongly believe that participants were not able to follow along all the emotions presented on the 22 seconds video and arrange them in the correct order. It is possible that, for instance, participants arranged fear instead of relief, and although they recognized the emotions correctly, they failed to report the right order. Perhaps it would have been helpful to label the emotions with numbers. By doing so, participants would have recognized exactly the section on the video they should pay special attention. Although this could have been helpful to improve the comprehension of the task, this would have been also a help for participants in their task of recognizing the interactive transitions of emotions. We wanted to

avoid this kind of aid during the task, because the goal was precisely to test the interactive change of emotions.

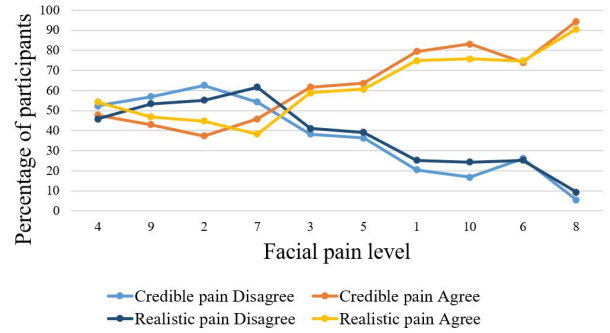


Figure 2: Percentage of participants rating realism and believability of facial pain expressions.

4 DEVELOPMENT SETUP

Based on the results of the focus group, we designed a series of emotional reactions guided by the FACS [11]. On a conceptual level, we identified a series of actions that paramedics and nurses perform, which should automatically trigger an appropriate emotional response, conforming an interaction concept that was partially evaluated with the online study. The final evaluation of the conception and its development should be evaluated in an upcoming study with end users, namely paramedic students.

4.1 Interaction Concept

In order to create an interaction concept it is necessary to establish which interactions should be developed. The concept of the VR prototype should include interactive elements that involve touch as well as the emotional reactions of the patient. These interactions are guided by the medical treatment, such as airway management, applying analgesics or bandages, and touching patients to comfort them. The decision of which actions and reactions need to be integrated into the VR prototype were conceptualized between the medical experts, designers and user experience researchers during and after the focus group (table 2). The facial expressions included in the concept are pain, fear and relief. Although other emotional reactions are possible, these were identified as the most relevant. Many of the reactions are automatically triggered after performing a concrete step. For example, injecting the patient with a syringe causes fear; removing the patient's clothes causes shame; touching a burnt area

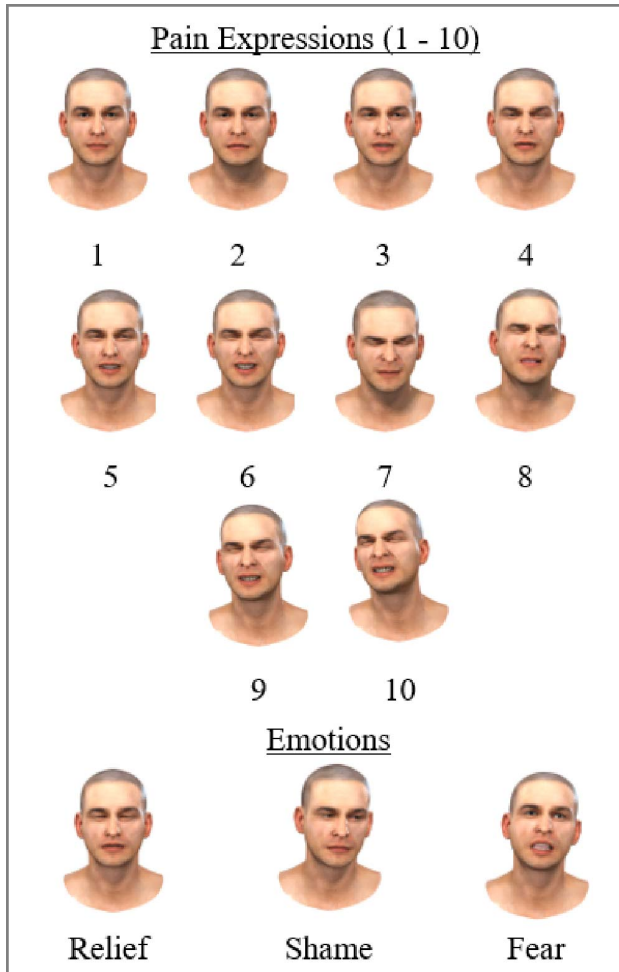


Figure 3: Facial expressions presented to the participants on Study 3.

(accidentally) causes brief severe pain, and touching an area that was not compromised causes brief relief; similarly administration of morphin and a relatively long eye contact with patient during communication also cause relief. Regarding the different pain reactions, the experts of the focus group mentioned that these are a fundamental part of the symptomatic, which in such simulations are usually adjusted directly by the trainer. Therefore, this interaction concept places the adjustments of pain expressions as a function in the trainer menu that allows trainers to adjust the pain responses of the virtual patient from 1 to 10. Nevertheless, we decided to program some of the pain reactions so that they appear automatically after a particular action (see application of bandages, pain management and touch on wound on table 2)

4.2 Design of facial expressions

Integrating such complex facial expressions in VR with similar detail as in the main study (figure 3) is a great challenge. Developing a rigged 3D model with textures that recreates a realistic human being with wrinkles that precisely emphasize on the respective emotion requires detailed work. Hence, in order to develop a first prototype, we used the rigged 3D model which is available from the VR training (study 1) and adapted it to a photogrammetric model of the simulation manikin using 3ds Max [6]. In this sense, improving an existing

Table 2: Interaction concept

Action	Emotional reaction
Remove clothes	Fear, strong pain, shame
Heat management	Fear, relief
Application of bandages	Fear, strong pain, relief
Pain management	Fear, Relief, mild pain
Constant communication	Relief, eye contact
Touch on shoulder	Relief, eye contact
Touch on wound	Strong pain

system limited some creative liberties usually available when developing a character from scratch. For instance, the proportions and appearance of the virtual character were already determined. In our particular case, the virtual character did not have all the necessary rigs to design emotions based on the FACS. Therefore, to create the facial expressions, we designed 4 different blendshapes. According to the FACS, eyes and mouth are the facial areas to which we should pay special attention. High pain levels are related to higher activation of muscles on these areas. The blendshapes were adjusted accordingly and overlaid in the application depending on the interaction. Thus, we were able to rework the mesh of the model as far as possible, reaching relatively accurate measurements with minor deviations from the original. Although it is quite challenging to fine-tune the pain and emotional expressions in detail, specially in a realistic VR simulation, we were able to integrate the most relevant aspects addressed on both studies. However, it is important to include auditive feedback to provide a holistic experience that includes visual and auditive elements. The medical experts of the focus group highlighted the relevance of the sounds (e.g. screams, moans) that patients make. By using Soundsnap [13], we chose suitable sounds and processed them into audio snippets, which should produce a holistic user experience that includes different perception modalities in a next development phase and prototype.

5 DISCUSSION

This paper presents the development of an interaction concept for emotions in a VR training for paramedics. Two studies with qualitative and quantitative methodologies guided the development process. The design of realistic emotions that are expected during the treatment of a patient is a promising way of improving user experience, since these features have increased empathy [7] and the number of interactions [14] in similar systems.

Based on observations of trainers during the focus group we created an interaction concept for the VR prototype, i.e. that we identified what actions performed by trainees on VR lead to an emotional reaction on the virtual patient. Furthermore, we identified which emotions and how many pain reactions are needed for the development phase. Touch is another relevant aspect when treating a patient. By allowing paramedic students to gently comfort the virtual patient by a touch on the shoulder, they gain experience that they could only achieve in real practice or with a patient-actor, which might be complicated to engage.

The results of the online study showed how the frequent exposure to the facial expressions increases their credibility and the realistic perception of participants. The learning effect that was observed on the first task indicates that trainees can quickly adapt and get used to the pain expressions, perceiving them as more realistic after a relatively short exposure. Furthermore, with the second task, we confirmed that our design, based on the FACS, generates an accurate depiction of the pain level shown by a virtual character. Although with this study was impossible to generate the fully immersive environment usually generated in VR, participants were still able to differentiate between very detailed and subtle changes on the pain

expressions. It is also relevant to mention that the study design presented certain problems in the execution, which indicates some limitations that compels us to interpret the results very carefully. However, the preparation of this study, which consisted on the design of realistic pain and emotional expressions, encourages us to recreate these emotions with such details in VR as well. We infer that with the integration of auditive feedback (e.g. screams and moans) within a narrative where actions lead to realistic emotions, the user experience will be improved. However, this needs to be tested in an evaluation with paramedic students.

6 CONCLUSIONS AND FUTURE WORK

We present an interaction concept for the emotional response of a patient in a VR medical training setting. One of the most relevant goals of this paper is to show how with a user-centered perspective and interdisciplinary research that includes medicine, education, interaction design, and computer science, we can develop systems that directly address scientific, educational and user requirements. In order to really observe the effects of our development, we are planning on conducting a user study with paramedic students to observe how they react to the emotional reactions of the patient. With a mixed methods approach we will assess relevant aspects around VR training, such as usability of the prototype, presence, empathy, and learning effects.

ACKNOWLEDGMENTS

This work is part of the project ViTAWiN (vitawin.info) which is financed by the German Federal Ministry of Education and Research under the grant 01PV18006.

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